

```
In [ ]: %pip install opencv-python
```

```
In [5]: %pip install matplotlib pillow
```

```
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp313-cp313-win_amd64.whl.metadata (80 kB)
Collecting pillow
  Using cached pillow-12.3.0-cp313-cp313-win_amd64.whl.metadata (9.3 kB)
Collecting contourpy≥1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp313-cp313-win_amd64.whl.metadata (3.8 kB)
Collecting cyclor≥0.10 (from matplotlib)
  Using cached cyclor-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools≥4.28.2 (from matplotlib)
  Downloading fonttools-4.65.0-cp313-cp313-win_amd64.whl.metadata (127 kB)
Collecting kiwisolver≥1.3.1 (from matplotlib)
  Using cached kiwisolver-1.5.1-cp313-cp313-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: numpy≥1.25 in
d:\programming_apps\miniconda3\lib\site-packages (from matplotlib) (2.5.3)
Requirement already satisfied: packaging≥20.0 in
d:\programming_apps\miniconda3\lib\site-packages (from matplotlib) (25.0)
Collecting pyparsing≥3 (from matplotlib)
  Using cached pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil≥2.7 in
d:\programming_apps\miniconda3\lib\site-packages (from matplotlib)
(2.9.0.post0)
Requirement already satisfied: six≥1.5 in
d:\programming_apps\miniconda3\lib\site-packages (from python-dateutil≥2.7
→matplotlib) (1.17.0)
Downloading matplotlib-3.11.2-cp313-cp313-win_amd64.whl (9.3 MB)
----- 0.0/9.3 MB ? eta -:-:-
- ----- 0.3/9.3 MB ? eta -:-:-
-- ----- 0.5/9.3 MB 1.5 MB/s eta 0:00:06
--- ----- 0.8/9.3 MB 1.4 MB/s eta 0:00:07
---- ----- 0.8/9.3 MB 1.4 MB/s eta 0:00:07
----- 1.3/9.3 MB 1.2 MB/s eta 0:00:07
----- 1.3/9.3 MB 1.2 MB/s eta 0:00:07
----- 1.6/9.3 MB 1.2 MB/s eta 0:00:07
----- 1.8/9.3 MB 1.2 MB/s eta 0:00:07
----- 2.1/9.3 MB 1.2 MB/s eta 0:00:07
----- 2.4/9.3 MB 1.2 MB/s eta 0:00:06
----- 2.6/9.3 MB 1.2 MB/s eta 0:00:06
----- 2.9/9.3 MB 1.2 MB/s eta 0:00:06
----- 3.1/9.3 MB 1.2 MB/s eta 0:00:06
----- 3.4/9.3 MB 1.2 MB/s eta 0:00:05
----- 3.7/9.3 MB 1.2 MB/s eta 0:00:05
----- 4.2/9.3 MB 1.2 MB/s eta 0:00:05
----- 4.5/9.3 MB 1.2 MB/s eta 0:00:04
----- 4.7/9.3 MB 1.2 MB/s eta 0:00:04
----- 5.0/9.3 MB 1.2 MB/s eta 0:00:04
----- 5.2/9.3 MB 1.2 MB/s eta 0:00:04
----- 5.5/9.3 MB 1.3 MB/s eta 0:00:04
----- 5.8/9.3 MB 1.3 MB/s eta 0:00:03
----- 6.0/9.3 MB 1.3 MB/s eta 0:00:03
----- 6.3/9.3 MB 1.3 MB/s eta 0:00:03
----- 6.6/9.3 MB 1.3 MB/s eta 0:00:03
```

```
----- 6.8/9.3 MB 1.3 MB/s eta 0:00:02
----- 7.3/9.3 MB 1.3 MB/s eta 0:00:02
----- 7.6/9.3 MB 1.3 MB/s eta 0:00:02
----- 7.9/9.3 MB 1.3 MB/s eta 0:00:02
----- 8.1/9.3 MB 1.3 MB/s eta 0:00:01
----- 8.4/9.3 MB 1.3 MB/s eta 0:00:01
----- 8.7/9.3 MB 1.3 MB/s eta 0:00:01
----- 9.2/9.3 MB 1.3 MB/s eta 0:00:01
----- 9.3/9.3 MB 1.3 MB/s eta 0:00:00
Using cached pillow-12.3.0-cp313-cp313-win_amd64.whl (7.2 MB)
Downloading contourpy-1.4.0-cp313-cp313-win_amd64.whl (234 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.65.0-cp313-cp313-win_amd64.whl (2.5 MB)
----- 0.0/2.5 MB ? eta -:--:--
----- 0.3/2.5 MB ? eta -:--:--
----- 0.5/2.5 MB 1.2 MB/s eta 0:00:02
----- 0.8/2.5 MB 1.3 MB/s eta 0:00:02
----- 1.0/2.5 MB 1.4 MB/s eta 0:00:02
----- 1.3/2.5 MB 1.4 MB/s eta 0:00:01
----- 1.8/2.5 MB 1.4 MB/s eta 0:00:01
----- 2.1/2.5 MB 1.4 MB/s eta 0:00:01
----- 2.4/2.5 MB 1.4 MB/s eta 0:00:01
----- 2.5/2.5 MB 1.4 MB/s eta 0:00:00
Using cached kiwisolver-1.5.1-cp313-cp313-win_amd64.whl (70 kB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools,
cycler, contourpy, matplotlib
```

```
----- 0/7 [pyparsing]
----- 0/7 [pyparsing]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
```

4/10

5/10

```

----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 6/7 [matplotlib]
----- 7/7 [matplotlib]

```

Successfully installed contourpy-1.4.0 cyclor-0.12.1 fonttools-4.65.0
 kiwisolver-1.5.1 matplotlib-3.11.2 pillow-12.3.0 pyparsing-3.3.2
 Note: you may need to restart the kernel to use updated packages.

```

In [11]: import cv2
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

image_path = r"C:\Lab3\Zebra_image.jpeg"
original_image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

def sample_image(image, factor):
    height, width = image.shape[:2]
    return cv2.resize(image, (width // factor, height // factor),
interpolation=cv2.INTER_NEAREST)

def quantize_image(image, levels):
    step = 256 // levels
    quantized_image = np.floor(image / step) * step
    return quantized_image.astype(np.uint8)

def plot_images(original, sampled, quantized):
    plt.figure(figsize=(12, 4))
    plt.subplot(1, 3, 1)
    plt.imshow(original, cmap='gray')
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(sampled, cmap='gray')
    plt.title('Sampled Image')
    plt.axis('off')

    plt.subplot(1, 3, 3)
    plt.imshow(quantized, cmap='gray')

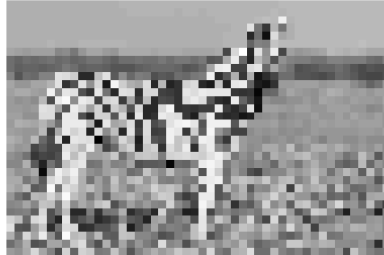
```

```
plt.title('Quantized Image')  
plt.axis('off')  
  
plt.tight_layout()  
plt.show()  
  
sampled_image = sample_image(original_image, 14)  
quantized_image = quantize_image(original_image, 9)  
  
plot_images(original_image, sampled_image, quantized_image)
```

Original Image



Sampled Image



Quantized Image



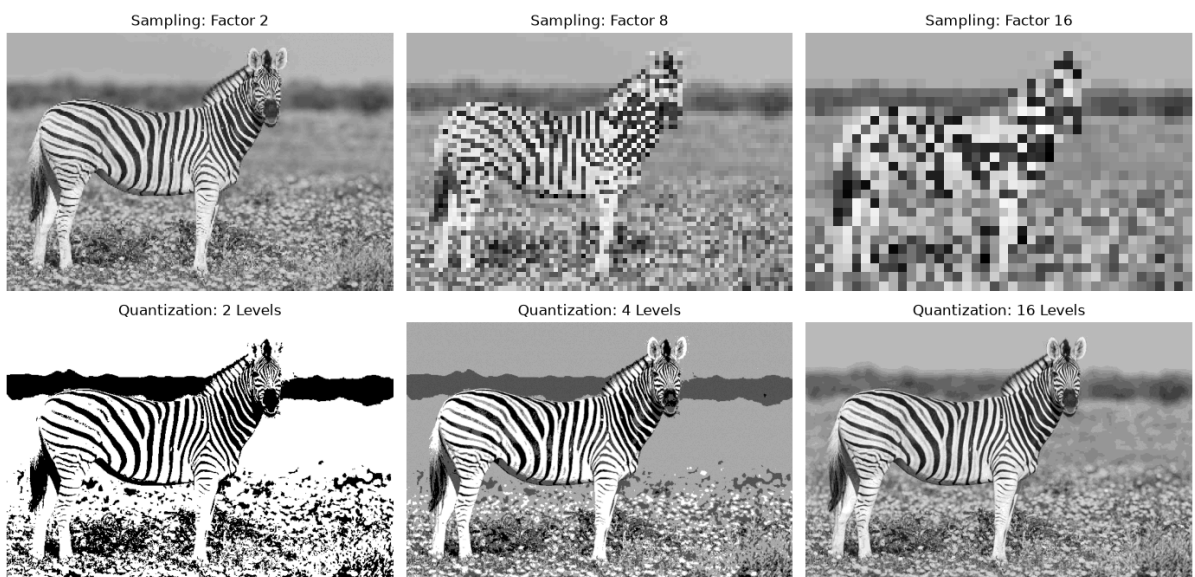
```
In [12]: factors = [2, 8, 16]
levels = [2, 4, 16]

plt.figure(figsize=(14, 7))

for i, factor in enumerate(factors, 1):
    downsampled = sample_image(original_image, factor)
    plt.subplot(2, 3, i)
    plt.imshow(downsampled, cmap='gray')
    plt.title(f'Sampling: Factor {factor}')
    plt.axis('off')

for i, level in enumerate(levels, 4):
    quantized = quantize_image(original_image, level)
    plt.subplot(2, 3, i)
    plt.imshow(quantized, cmap='gray')
    plt.title(f'Quantization: {level} Levels')
    plt.axis('off')

plt.tight_layout()
plt.show()
```



```
In [13]: img1 = cv2.resize(original_image, (400, 400))
img2 = cv2.flip(img1, 1)

im1arr = np.asarray(img1)
im2arr = np.asarray(img2)

addition = im1arr + im2arr
union = im1arr | im2arr

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(addition, cmap='gray')
plt.title('Addition')
plt.axis('off')

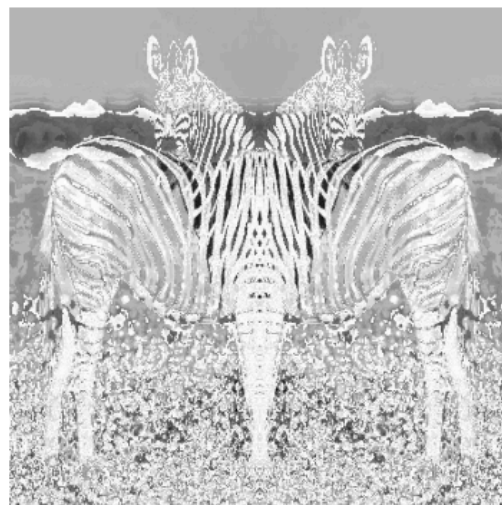
plt.subplot(1, 2, 2)
plt.imshow(union, cmap='gray')
plt.title('Union')
plt.axis('off')

plt.tight_layout()
plt.show()
```

Addition



Union



```
In [14]: im1_int = img1.astype(np.int16)
im2_int = img2.astype(np.int16)

subtracted = np.clip(im1_int - im2_int, 0, 255).astype(np.uint8)
add_175 = np.clip(im1_int + 175, 0, 255).astype(np.uint8)
set_diff = img1 & (~img2)
sym_diff = img1 ^ img2
intersection = img1 & img2

outputs = [
    ("1. Subtraction", subtracted),
    ("2. Add Constant (175)", add_175),
    ("3. Set Difference", set_diff),
    ("4. Symmetric Difference", sym_diff),
    ("5. Intersection", intersection)
]

plt.figure(figsize=(15, 6))
for idx, (title, img_data) in enumerate(outputs, 1):
    plt.subplot(1, 5, idx)
    plt.imshow(img_data, cmap='gray')
    plt.title(title)
    plt.axis('off')

plt.tight_layout()
plt.show()
```

